Before vs After Analysis

Capstone 14: AI Accessibility Study Aids | NCERT Class 10 Science, Ch. 11, Section 11.1 | Prepared by Divyalakshmi.G

Question: Did the AI-assisted versions (simplified text, audio, mind-map) make the same content easier to access than the original, and what was lost or risked on the way?

Method: The original is the NCERT Section 11.1 text, including Example 11.1. The "after" versions are the final merged simplified text, the final audio and the final mind-map. Text measures were computed by script from both texts. Peer feedback comes from three real Google Form responses.

1. Simplified text: original vs final

Measure	Original NCERT	Final simplified	Change
Words	525	344	181 fewer
Sentences	30	41	More, shorter sentences
Average sentence length	17.5 words	8.4 words	9.1 words shorter
Longest sentence	33 words	15 words	18 words shorter
Sentences of 15 words or more	16 of 30 (53%)	1 of 41	Target was under 15 words
Flesch Reading Ease (higher is easier)	59	77	+18
Flesch-Kincaid grade level	9.3	4.6	4.7 grades lower
Silent reading time (200 words/min)	about 2.6 min	about 1.7 min	Shorter, but see limits

Readability scores are rough estimates from a simple script. Formulas and numbers count as words, and the syllable count is approximate, so treat the grade level as an indication, not an exact value. The sentence-length figures are the most reliable.

What improved: shorter sentences, plain-word explanations ("Water flows in a river. This is a water current."), the circuit and torch example broken into steps, the reason behind the direction rule stated clearly, and key terms listed at the end. **What the simplified text kept:** every formula (I = Q/t, 1 A = 1 C / 1 s, Q = It), every unit and power of ten, and the worked example. **What was dropped:** the pointer to Figure 11.1 and the phrase "schematic diagram".

2. Audio: before vs after

Measure	Before (text only)	After (audio)
How a student gets the content	Must read text with sight	Can listen without reading
Time	about 2.6 min silent reading of the original	149.9 seconds (about 2 min 30 s) of audio
Numbers and formulas	Printed symbols (6×10^{18}, 10^{-19})	Spoken in words ("six times ten to the power eighteen"); no misreads found on listening

Measure	Before (text only)	After (audio)
Support for the listener	None	Section signposts ("Next, the direction of current"), variables defined aloud (I, Q, t), 3-point recap
Known risk		A listener cannot re-read quickly. Powers of ten are harder by ear.
Peer response		3 of 3 said the voice was clear. 2 rated text-to-voice Very useful, 1 Helpful.

The audio is not much faster than silent reading. Its benefit is access: students with visual impairment, dyslexia or tiredness from reading can learn without reading text.

3. Visual summary: before vs after

Measure	Before	After (mind-map)
Visual support in the original	One circuit diagram (Figure 11.1)	6 branches, 18 points, doodles that illustrate each idea
Length of each point	Full sentences	Average 6.0 words per point; 2 of 18 points over 8 words (both formula lines)
Direction of current	One dense paragraph	Two labelled arrows: electron flow and current flow, in words as well as colour
Peer response		3 of 3 said the mind-map was useful
Known errors left		1: the switch in branch 1 is drawn open while the bulb glows (still to be redrawn as closed)

4. Raw AI output vs final version (accuracy)

Format	Raw tool outputs (issues logged)	Final version
Simplified text	ChatGPT: 10 items, 2 high (wrong definition of charge, broken exponents). NotebookLM: 8 items, none high. Gemini: 9 items, 2 high (wrong source tag, garbled exponents).	Known errors fixed using the NCERT text. Student to confirm by reading it against pages 171-172.
Audio script	All three tools spoke every number and exponent correctly. Issues were filler, added wording ("tiny") and one imprecise recap point (NotebookLM). NotebookLM needed 2 attempts.	Merged script, filler and tool artifacts removed. Audio listened to in full: no misreads.
Mind-map outline	ChatGPT: 8 issues. NotebookLM: 6. Gemini: 9 (dropped "nearly", incomplete example, missing units).	Merged outline restores all missing facts and the negative sign.

Every raw output needed human correction. No tool followed every rule given in the prompt.

5. Peer feedback: before vs after

Status: collected. Three real responses were received through a Google Form linked to a Google Sheet. Reviewers are shown as Peer 1, 2 and 3 and their names are not used. The form used labelled answers (for example Excellent, Useful, Satisfied), not 1-5 numbers, so results are shown as counts.

Question in the form	Peer 1	Peer 2	Peer 3	Summary
How easy was the tool to use?	Excellent	Good	Excellent	2 Excellent, 1 Good
Was the text-to-voice feature helpful?	Very useful	Helpful	Very useful	2 Very useful, 1 Helpful
Was the voice message clear and understandable?	Yes	Yes	Yes	3 of 3 Yes
Was the mind-map useful for understanding?	Useful	Useful	Useful	3 of 3 Useful
Overall, how satisfied are you with the project?	Satisfied	Satisfied	Satisfied	3 of 3 Satisfied
Any suggestions for improvement?	"Nothing"	(left blank)	"Nothing... All are good"	No improvements suggested

Item	Result
Mistakes or missing points reported by peers	None reported
Changes made because of peer feedback	None needed from the comments. The open switch in the mind-map was found during review of the picture, not by peers.
Which format helped most	Not asked in this form
Clarity of the simplified text	Not asked in this form (see limits)

6. Findings

- **Text:** average sentence length fell from 17.5 to 8.4 words, the longest sentence from 33 to 15 words, and the estimated grade level from 9.3 to 4.6. All formulas, units and the worked example were kept.
- **Audio and visual:** they add new ways to access the same content. The audio is about as long as reading, so its value is access, not speed.
- **Peers:** all 3 reviewers said the voice was clear, the mind-map was useful and they were satisfied overall. None reported a mistake or suggested an improvement.
- **Accuracy:** the simplified formats were only trustworthy after human checking against NCERT. Raw outputs contained wrong definitions, broken exponents, added facts and a wrong source tag.
- **Trade-off:** shorter text and shorter mind-map points are easier to read but drop detail, such as the reason for the direction rule in the mind-map and the pointer to Figure 11.1 in the text.

7. Limits of this analysis

- Readability scores are approximate and were computed on one section only.

- Only 3 peers responded, all gave positive answers and none left suggestions, so there is little feedback to act on. No peer noticed the open switch in the mind-map.
- The form used labelled answers, not 1-5 scores, so no averages can be calculated. It asked about the tool, voice and mind-map, but not directly about the clarity of the simplified text.
- The simplified text was checked against the NCERT source by line-by-line comparison; the student should confirm this by reading it once more.
- Results come from one section of one chapter and may differ for other subjects.